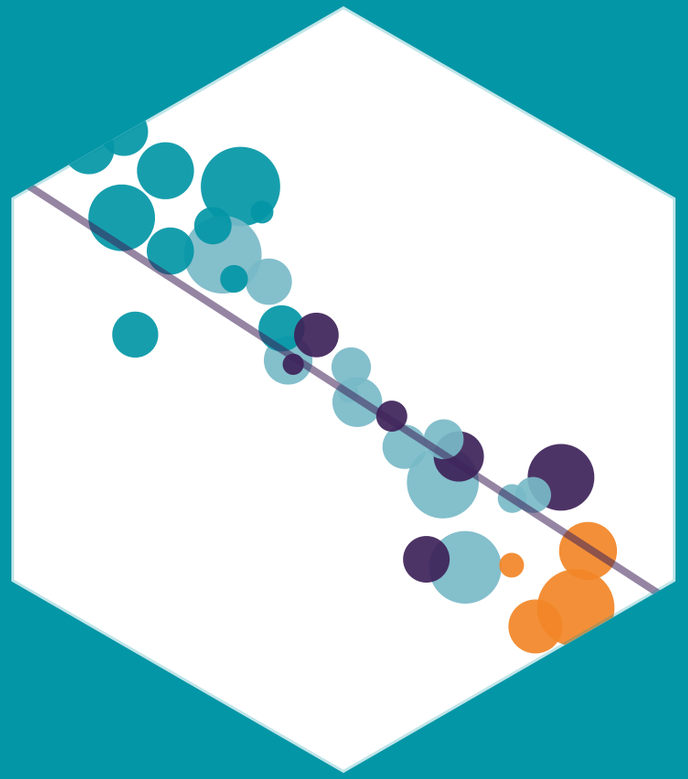
ECONOMETRICS

TA SESSION 6

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Planification — 2° Part

TA6 — RCTs

TA7 — Matching

TA8 — Instrumental Variables

📅 Hand in Assignment 6 by **18th Nov**

TA9 — Regression Discontinuity

📅 Hand in Assignment 7 by **27th Nov**

TA10 — Difference-in-Differences

📅 Hand in Assignment 8 by **2nd Dec**

Today

1. **Potential outcomes** — the fundamental problem of causality *(smoking & birthweight)*
2. **Randomization** — why it solves the problem
3. **An RCT in a ride-hailing app** — design, estimation, balance, covariates, heterogeneity

📓 Everything in **TA6.ipynb** (the data are simulated inside the notebook)

1 · Potential outcomes

Potential outcomes

$$D_i \in \{0,1\} \quad \text{treatment indicator}$$

$$Y_i(1) \quad \text{potential outcome } \textbf{with} \text{ treatment}$$

$$Y_i(0) \quad \text{potential outcome } \textbf{without} \text{ treatment}$$

$$\tau_i = Y_i(1) - Y_i(0) \quad \text{individual causal effect}$$

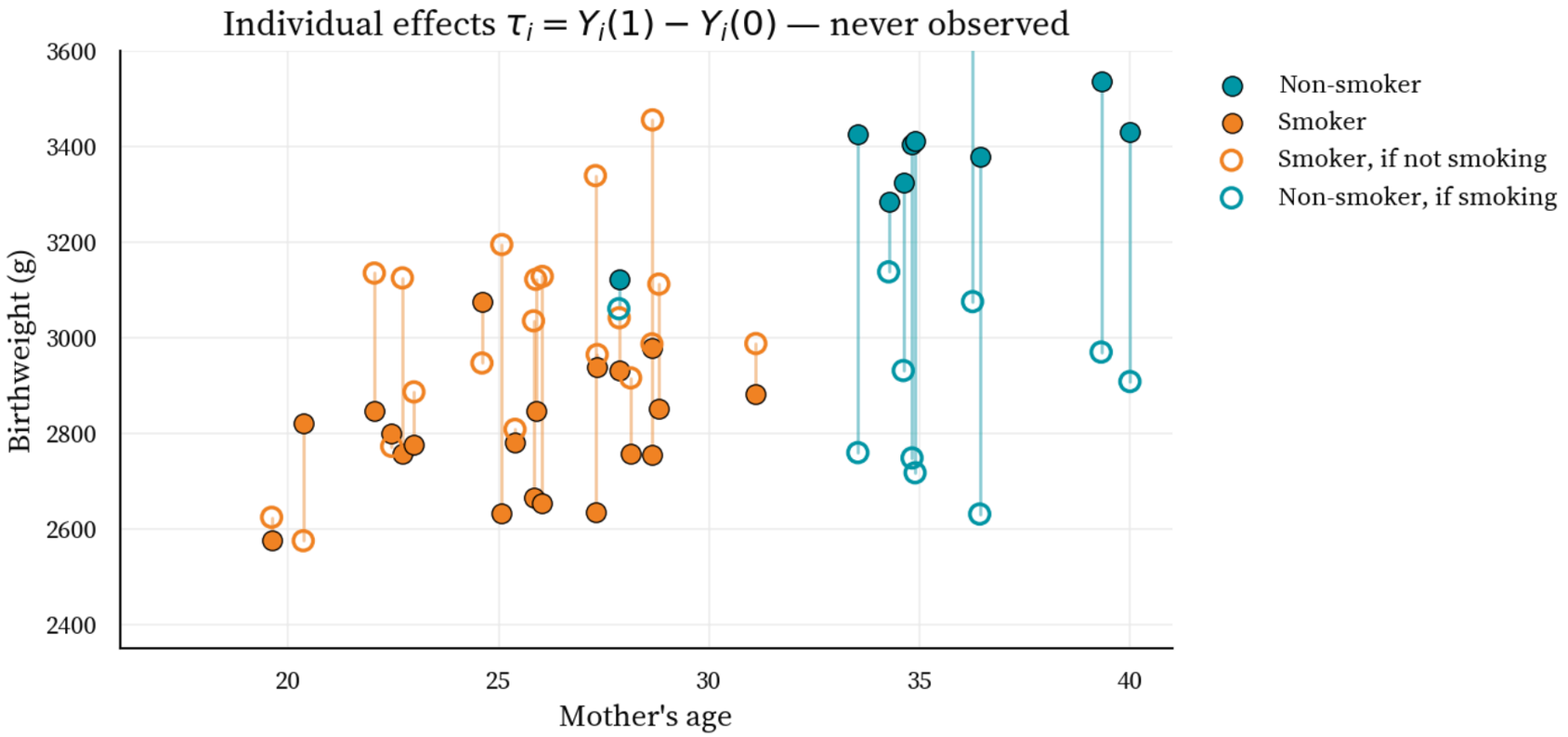
What we observe:

$$Y_i \;=\; D_i\,Y_i(1) + (1 - D_i)\,Y_i(0) \;=\; \begin{cases} Y_i(1) & \text{if } D_i = 1 \\ Y_i(0) & \text{if } D_i = 0 \end{cases}$$

The fundamental problem of causal inference

We **never** observe both $Y_i(1)$ and $Y_i(0)$ for the same person: causality is a **missing data** problem.

Example: does smoking during pregnancy reduce birthweight?



Each vertical segment is one τ_i. We only ever see **one end** of it.

Two averages of τ_i: ATE and ATT

$$\text{ATE} = E[Y(1) - Y(0)] \qquad \text{ATT} = E[Y(1) - Y(0) \mid D = 1]$$

- **ATE**: average effect over **everyone**. *"What if all mothers smoked vs. none?"*
- **ATT**: average effect over the **treated** only. *"How much lighter are smokers' babies because their mothers smoked?"*

What does the naive comparison of means estimate?

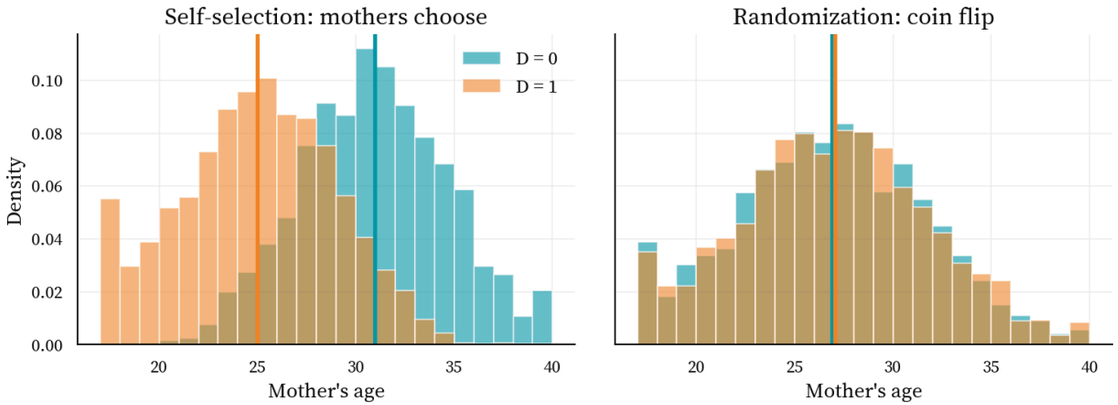
$$\underbrace{E[Y \mid D = 1] - E[Y \mid D = 0]}_{\text{naive comparison}} = \underbrace{E[Y(1) - Y(0) \mid D = 1]}_{\text{ATT}} + \underbrace{E[Y(0) \mid D = 1] - E[Y(0) \mid D = 0]}_{\text{selection bias}}$$

Age is a confounder: younger mothers smoke more **and** have lighter babies anyway ⇒ selection bias ≠ 0, so the naive comparison is neither the ATT nor the ATE.

2 · Randomization

Randomization kills selection bias

If D_i is assigned by a **coin flip**, it cannot depend on anything about the mother, observed or not:



$$\big(Y(1), Y(0)\big) \perp\!\!\!\perp D \implies \underbrace{E[Y(0) \mid D=1] - E[Y(0) \mid D=0]}_{\text{selection bias } = 0}$$

$$\implies E[Y \mid D=1] - E[Y \mid D=0] \;=\; \text{ATT} \;=\; \text{ATE}$$

Caveat

Here randomization is **not an option**: assigning pregnant women to smoke would be **unethical** (and infeasible). For questions like this we need **observational** methods: matching, IV, RDD, DiD (the next TAs).

3 · An RCT in a ride-hailing app

The case: a new, cheaper option in the app

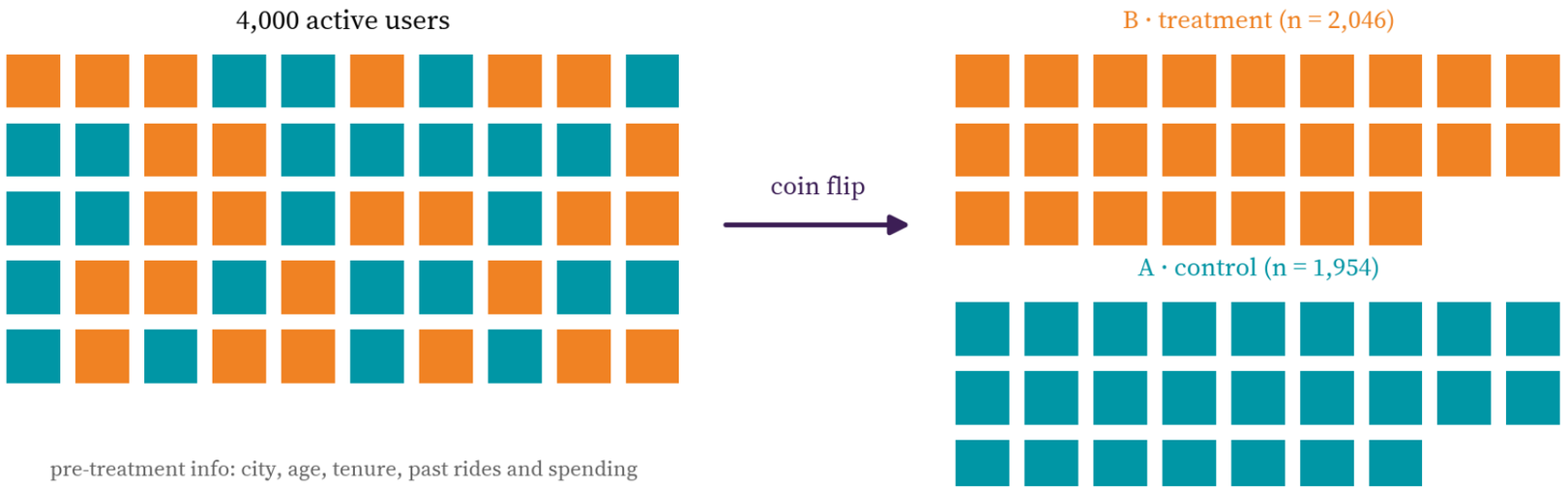
Ryde, a (fictional) ride-hailing app, wants to add **one** new option to the menu: **“Wait & Save”**: ~25% cheaper if you accept being picked up in 10–15 minutes.



Hypothesis: IF we add an option **“Wait & Save” THEN** price-sensitive users who today take the bus or walk will now request a ride bringing more **revenue**.

Designing the experiment

- **Outcome Y_i: revenue** from user i in the 4 weeks of the test.
- **Treatment $D_i = 1$** if user i sees app **B** (with "Wait & Save"), $D_i = 0$ if app **A** (current menu).
- **Units:** active users, each randomly assigned to A or B with probability ½ (a coin flip per user).



4 · Estimation

The data

We simulate the experiment (code in the notebook) for 4,000 users. Relevant variables:

- `revenue_post` : Y_i, revenue (€) from user i during the 4 weeks of the test
- `treat` : D_i, equal to 1 if app **B** (with "Wait & Save"), 0 if app **A**
- `city` , `age` , `tenure` (months on the app), `rides_pre` , `revenue_pre` (rides and € in the 4 weeks **before**): pre-treatment covariates X_i

```
1 app[["user_id", "city", "age", "tenure", "rides_pre", "revenue_pre", "treat", "revenue_post"]].head()
```

	user_id	city	age	tenure	rides_pre	revenue_pre	treat	revenue_post
0	1	Madrid	37	15	5	48.86	0	9.25
1	2	Madrid	33	35	7	83.38	0	114.59
2	3	London	35	9	6	117.30	1	115.93
3	4	Madrid	18	11	0	0.00	1	12.67
4	5	London	33	27	15	329.65	1	270.34

ATE = difference in means

Randomization ⇒ no selection bias ⇒ $\text{ATE} = E[Y \mid D=1] - E[Y \mid D=0]$, estimated by $\widehat{\text{ATE}} = \bar{Y}_1 - \bar{Y}_0$.

$$H_0: \; E[Y \mid D=1] = E[Y \mid D=0] \qquad H_a: \; E[Y \mid D=1] \neq E[Y \mid D=0]$$

$$t = \frac{\bar{Y}_1 - \bar{Y}_0}{\sqrt{s_1^2/n_1 + s_0^2/n_0}} \overset{H_0}{\approx} N(0,1)$$

Welch two-sample t-test (unequal variances): reject H_0 at 5% if $|t| > 1.96$.

```
y1 = app.loc[app.treat == 1, "revenue_post"]          # D = 1 (app B)
y0 = app.loc[app.treat == 0, "revenue_post"]          # D = 0 (app A)
print(f"mean(D=1) = {y1.mean():.2f}   mean(D=0) = {y0.mean():.2f}   difference = {y1.mean() - y0.mean():.2f}")
res = stats.ttest_ind(y1, y0, equal_var=False)          # Welch t-test
print(f"t = {res.statistic:.3f}   p-value = {res.pvalue:.4f}   df = {res.df:.1f}")
```

```
mean(D=1) = 60.88   mean(D=0) = 57.65   difference = 3.23
t = 1.612   p-value = 0.1070   df = 3996.9
```

$t = 1.61,\ p = 0.107 \Rightarrow$ we **cannot reject** H_0 at 5%: the estimate is positive, but too noisy.

… which is the same as a regression on D

$$\texttt{revenue_post}_i = \beta_0 + \beta_1 D_i + \varepsilon_i$$

OLS Regression Results

Dep. Variable:	revenue_post	R-squared:	0.001
Model:	OLS	Adj. R-squared:	0.000
Method:	Least Squares	F-statistic:	2.599
Date:	Mon, 09 Nov 2026	Prob (F-statistic):	0.107
Time:	21:08:30	Log-Likelihood:	-22276.
No. Observations:	4000	AIC:	4.456e+04
Df Residuals:	3998	BIC:	4.457e+04
Df Model:	1		
Covariance Type:	HC1		

	coef	*std err*	*z*	*P>|z|*	*[0.025*	*0.975]*
Intercept	57.6523	1.389	41.513	0.000	54.930	60.374
treat	3.2309	2.004	1.612	0.107	-0.697	7.159

Omnibus:	2005.560	Durbin-Watson:	2.049
Prob(Omnibus):	0.000	Jarque-Bera (JB):	14276.113
Skew:	2.303	Prob(JB):	0.00
Kurtosis:	11.028	Cond. No.	2.65

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Are treatment and control truly random?

If randomization worked, B and A look alike **before** the test. For each pre-treatment covariate X_k:

$$H_0: \; E[X_k \mid D=1] = E[X_k \mid D=0] \qquad t_k = \frac{\bar{X}_{k,1} - \bar{X}_{k,0}}{\sqrt{s^2_{k,1}/n_1 + s^2_{k,0}/n_0}}$$

X_k	$\bar{X}_{k,1}$	$\bar{X}_{k,0}$	Diff.	t_k	p
Age	33.767	33.912	-0.145	-0.53	0.60
Tenure (months)	19.747	20.278	-0.531	-0.88	0.38
Rides before	3.832	3.815	+0.018	0.14	0.89
Revenue before (€)	56.614	56.286	+0.328	0.15	0.88
City = Madrid (share)	0.298	0.295	+0.003	0.23	0.82
City = Barcelona (share)	0.297	0.303	-0.007	-0.47	0.64
City = Paris (share)	0.209	0.203	+0.006	0.47	0.64
City = London (share)	0.196	0.199	-0.003	-0.21	0.84

No difference is significant ✅ (with many tests, ~1 in 20 rejects at 5% by chance). Joint check, regressing D on all X: $F = 0.21$, $p = 0.98$.

Using covariates to estimate the ATE

$$\texttt{revenue_post}_i = \beta_0 + \beta_1 D_i + \gamma_1\,\texttt{age}_i + \gamma_2\,\texttt{tenure}_i + \gamma_3\,\texttt{revenue_pre}_i + \sum_c \delta_c\,\text{city}_{c,i} + \varepsilon_i$$

OLS Regression Results

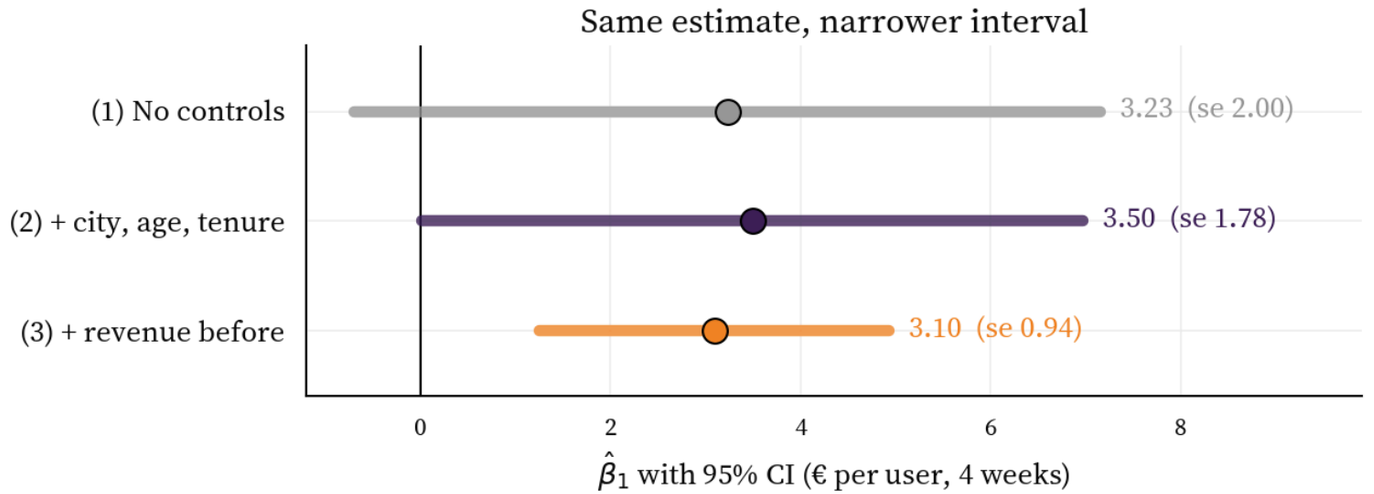
Dep. Variable:	revenue_post	R-squared:	0.781
Model:	OLS	Adj. R-squared:	0.780
Method:	Least Squares	F-statistic:	649.2
Date:	Mon, 09 Nov 2026	Prob (F-statistic):	0.00
Time:	21:08:30	Log-Likelihood:	-19242.
No. Observations:	4000	AIC:	3.850e+04
Df Residuals:	3992	BIC:	3.855e+04
Df Model:	7		
Covariance Type:	HC1		

	coef	*std err*	*z*	*P>|z|*	*[0.025*	*0.975]*
Intercept	5.2620	2.111	2.493	0.013	1.125	9.399
city[Barcelona]	2.3287	0.921	2.528	0.011	0.524	4.134
city[Paris]	8.1180	1.298	6.252	0.000	5.573	10.663
city[London]	15.8658	1.655	9.588	0.000	12.622	19.109
treat	3.0957	0.939	3.297	0.001	1.256	4.936
age	-0.0529	0.052	-1.008	0.313	-0.156	0.050
tenure	0.2283	0.030	7.722	0.000	0.170	0.286
revenue_pre	0.7825	0.014	57.607	0.000	0.756	0.809

Omnibus:	484.306	Durbin-Watson:	2.048
Prob(Omnibus):	0.000	Jarque-Bera (JB):	2317.945
Skew:	0.491	Prob(JB):	0.00
Kurtosis:	6.597	Cond. No.	460.

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Covariates shrink the CI, not the bias



	(1) No controls	(2) + city, age, tenure	(3) + revenue before
R^2	0.001	0.218	0.781
p-value of $\hat{\beta}_1$	0.107	0.049	0.001

Heterogeneous treatment effects: by city

$$\texttt{revenue_post}_i = \beta_0 + \beta_1 D_i + \sum_{c \neq \text{Madrid}} \delta_c \, \text{city}_{c,i} + \sum_{c \neq \text{Madrid}} \theta_c \, (D_i \times \text{city}_{c,i}) + \varepsilon_i$$

OLS Regression Results

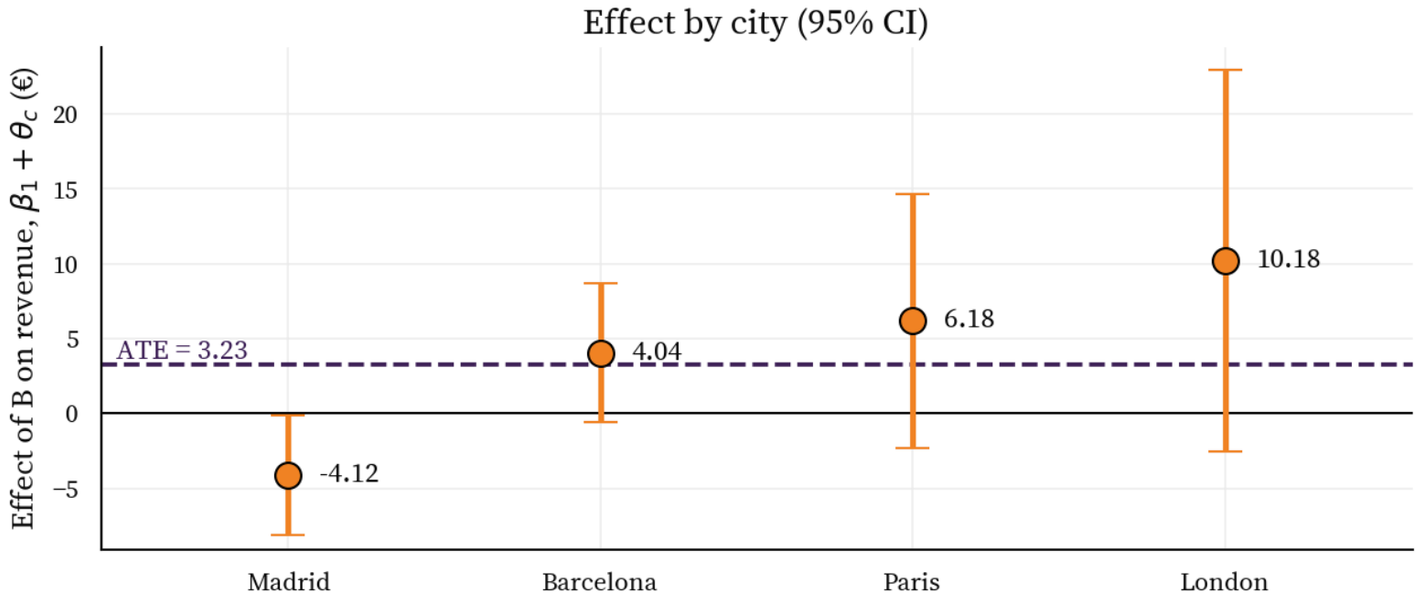
Dep. Variable:	revenue_post	R-squared:	0.176
Model:	OLS	Adj. R-squared:	0.175
Method:	Least Squares	F-statistic:	84.81
Date:	Mon, 09 Nov 2026	Prob (F-statistic):	2.13e-115
Time:	21:08:31	Log-Likelihood:	-21889.
No. Observations:	4000	AIC:	4.379e+04
Df Residuals:	3992	BIC:	4.385e+04
Df Model:	7		
Covariance Type:	HC1		

	coef	*std err*	*z*	*P>|z|*	*[0.025*	*0.975]*
Intercept	37.3973	1.534	24.372	0.000	34.390	40.405
city[Barcelona]	5.3419	2.245	2.380	0.017	0.942	9.742
city[Paris]	29.0204	3.455	8.400	0.000	22.249	35.791
city[London]	64.0574	4.710	13.601	0.000	54.827	73.288
treat	-4.1250	2.024	-2.038	0.042	-8.092	-0.158
treat × city[Barcelona]	8.1618	3.117	2.619	0.009	2.053	14.271
treat × city[Paris]	10.3056	4.790	2.151	0.031	0.917	19.694
treat × city[London]	14.3082	6.819	2.098	0.036	0.943	27.673

Omnibus:	1664.592	Durbin-Watson:	2.044
Prob(Omnibus):	0.000	Jarque-Bera (JB):	10530.678
Skew:	1.869	Prob(JB):	0.00
Kurtosis:	10.015	Cond. No.	11.9

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Heterogeneous treatment effects: by city



Joint test $H_0 : \theta_{BCN} = \theta_{PAR} = \theta_{LON} = 0$: $F = 3.79$, $p = 0.010$ → we **reject** that the effect is the same in every city. Revenue rises most in **London** (€10.18) and **Paris** (€6.18). In **Madrid** the estimate is **negative**: users mostly switch to the cheaper option instead of riding more, so revenue falls. Randomization holds within each city, so each city effect is causal (but pre-specify subgroups: slice the data 20 ways and one will look significant by chance).

Appendix

Take-up: offering ≠ using

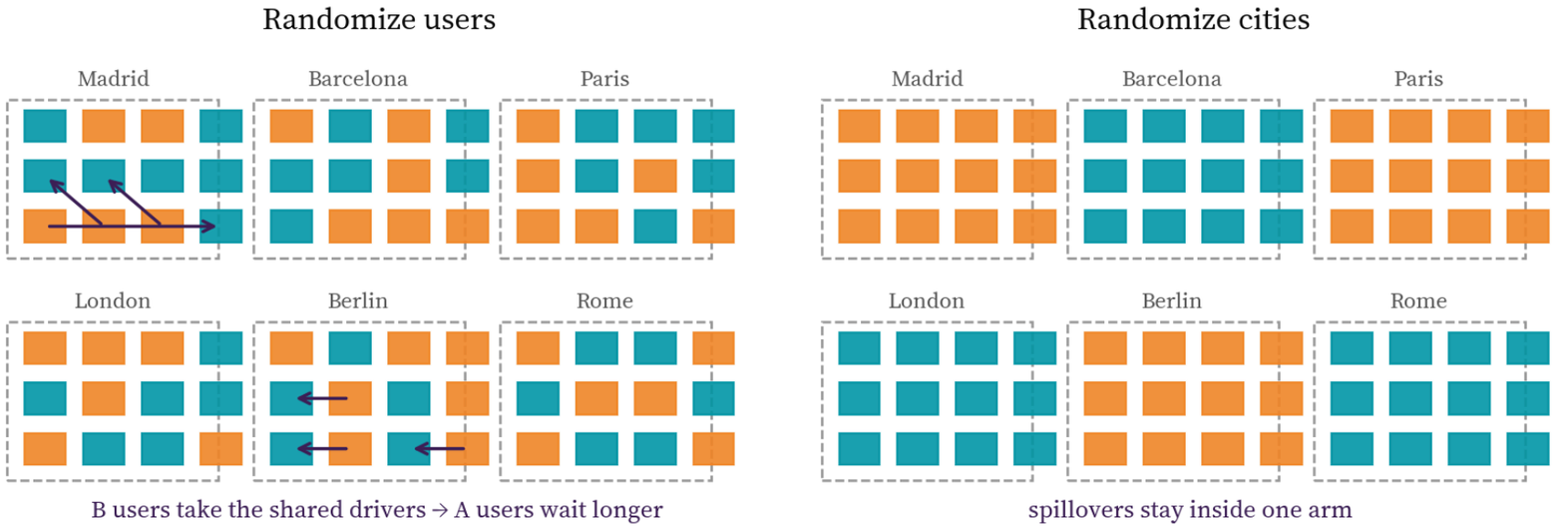
We can **offer** "Wait & Save" (app B), but we cannot force users to **choose** it.

Users in B	N	Revenue **before** (€)	Revenue during (€)
Did not choose it	1,239	40.71	45.25
Chose it	807	81.04	84.89

- Only **39%** of B users chose the new option.
- $\hat{\beta}_1$ = €3.10 is the effect of **being offered** the option: the **intention-to-treat (ITT)**. Still causal, and the relevant number for the launch decision.
- ❌ Don't compare choosers vs non-choosers: choosers already spent much more **before** the test → self-selection, the same problem as smokers.
- Effect on those who actually chose it ≈ ITT / take-up = 3.10 / 0.39 = **€7.85** (more on this with IV, TA8).

When SUTVA fails, change the unit of randomization

SUTVA: my outcome depends only on **my own** treatment, not on other users' treatment.



How the spillover happens. Users in the same city share the **same pool of drivers**. B users request extra (cheap) rides → fewer free drivers → **A users** wait longer, see surge prices, or give up → A's revenue falls **because of B**. So $\bar{Y}_0$ no longer estimates $E[Y(0)]$, and $\bar{Y}_1 - \bar{Y}_0$ **overstates** the effect of launching the option for everyone.

Fix: randomize at the level where the interaction happens: the **city** (the driver market). All users in a city see the same app, so spillovers stay *within* one arm; **cluster SEs by city**. Cost: few units (cities) → much less power. In practice firms use many cities, or **switchback** designs (randomize city × time slot).

Thanks!